

Selected-run evidence status: failed

## Geometry

No matching geometry image or recorded mesh is available for this run.

## Mesh

### Mesh overview and local detail

No selected-run setup values are recorded.

Mesh overview unavailable: Mesh exceeds the bounded report renderer capacity

## Setup

Recorded selected-case model

|                                 |            |               |
|---------------------------------|------------|---------------|
| Initial impact velocity         | 50         | m/s           |
| Requested simulation duration   | 8          | ms            |
| Foam depth                      | 203.2      | mm            |
| Foam width                      | 41.275     | mm            |
| Foam height                     | 152.4      | mm            |
| Foam element size               | 10         | mm            |
| Initial contact gap             | 0.1        | mm            |
| Contact friction coefficient    | 0.1        | dimensionless |
| Foam curve scale                | 1          | dimensionless |
| Time-step scale                 | 0.6        | dimensionless |
| Requested state-output interval | 0.05       | ms            |
| Startup mass-scaling DT2MS      | -1.5e-07   | s             |
| Termination time-step ratio     | 0.01       | dimensionless |
| Printed parts mass              | 0.5620892  | kg            |
| Hardware mass                   | 0.03400217 | kg            |
| Motor mass                      | 0.16       | kg            |

|                        |           |    |
|------------------------|-----------|----|
| Moving mass            | 0.7560913 | kg |
| Initial kinetic energy | 945.1142  | J  |

**Recorded material, contact and boundary assumptions**

- Unchanged prior clamshell fixture and synthetic 54.46 kg/m3 MAT057 foam; not the proposed 30 kg/m3 product.
- Three independent printed solids; body-to-closure frictional contact plus eight ideal bolt collars; no slip/preload/failure capacity.
- Eight CAD generic steel bolt/nut envelope masses distributed on the collar nodes.
- Two 80 g motor masses distributed on the original motor bores; no casing stiffness.
- PA12-CF10 isotropic linear elastic screen; not printed anisotropic strength qualification.
- Ideal transverse cable-bore guidance leaves axial Z free; no cable compliance, tension or clearance contact.
- Two stationary aluminum spreaders, 0.1 mm initial gap; fixed foam rear Z, free foam sides.
- No carrier, motor thrust, element failure, fracture or damage.

**Solve**

|                     |        |
|---------------------|--------|
| Recorded run status | failed |
|---------------------|--------|

**Separate solve charts**

## Contact force resultant

No extracted history for this result in the selected run.

Contact force resultant. No extracted source available.

## Mass-weighted forward velocity

No extracted history for this result in the selected run.

Mass-weighted forward velocity. No extracted source available.

## Moving assembly travel

No extracted history for this result in the selected run.

Moving assembly travel. No extracted source available.

## Foam compression

No extracted history for this result in the selected run.

Foam compression. No extracted source available.

## Regional P99 von Mises stress

No extracted history for this result in the selected run.

Regional P99 von Mises stress. No extracted source available.

## Global energy

No extracted history for this result in the selected run.

Global energy. No extracted source available.

## Results

# Run 031 - impact evidence

**Evidence status:** failed. This report describes only extracted evidence from the selected run. Numerical checks alone do not establish physical qualification.

## Geometry and model setup

| Recorded model item | Value          |
|---------------------|----------------|
| kind                | impact         |
| assembly kind       | local_closures |
| printed kg          | 0.562089       |
| hardware kg         | 0.0340022      |
| motors kg           | 0.16           |
| moving kg           | 0.756091       |
| initial ke J        | 945.114        |
| spreader each kg    | 0.135871       |

## Solved mesh

Recorded mesh source: baseline\_3mm. Mesh and timestep convergence are not established by these histories.

## Motion and stopping event

Signed forward velocity, travel and foam compression are measured histories. Negative forward velocity denotes rebound; a partial history cannot establish the complete stopping event.

| Measured result | Value | Unit |
|-----------------|-------|------|
|-----------------|-------|------|

## Regional stress and energy

Regional P99 values are elastic demand measures across saved states, not printed-part strength predictions. Force peaks and percentile peaks can occur at different times.

## Numerical checks and qualification

No selected-run numerical check summary is available.

No extracted results are available for this run yet.

## Images, animations and source evidence

- execution.json
- model.json
- setup\_audit.json

Generated from selected-run extracted evidence; no authored report is available. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.